Peer Review File

Editorial Note: Parts of this Peer Review File have been redacted as indicated to remove third-party material from other papers.

Manuscript Title: Singlet and triplet Cooper pair splitting in hybrid superconducting nanowires **Reviewer Comments & Author Rebuttals**

Reviewer Reports on the Initial Version:

Referees' comments:

Referee #1 (Remarks to the Author):

The paper by Wang and co-workers contains the experimental investigation of Cooper pair splitting with a spin filtering output. A point of interest is the transformation of the standard spin singlet pairing and its splitting into equal spin triplet pairing through spin-orbit interaction. I think the assessment of this work is best split into three different parts: (1) the quality of the obtained results, (2) the analysis and presentation of the results, and (3) the significance of the results.

Item (1) is of high standard. The obtained linewidths, the stability diagrams, seeming lack of cross-talk between the dots show all an excellent sample quality and experimental access. The main results presented are supported by several well chosen and indeed required extended data plots. There isn't much I can add, this is state-of-the-art. On the other hand, items (2) and (3) have shortcomings. The analysis and presentation, item (2), have made only an incomplete transition from what is clear to the authors to what the reader will need to know and when. I leave the issue open that the text would be inaccessible to anyone not fully familiar with superconducting nanostructures, and I start from considering that e.g. the provided description of CAR, ECT etc. is sufficient. Yet I find quite a number of important statements made in a way that implies more that one should trust the authors rather than allowing the readers to verify the conclusions themselves. Let me start with Fig. 2. From the sketches in panels b and c one can indeed infer the shown signals and their signs on the diagonals of the following panels. But from the sketches one would also assume that a convincing evidence for the cross-over from ECT to CAR should show up from the changing signal when continuously lowering of V_{LD} at fixed V_{RD}. If one looks at the voltage ranges, however, there is a discontinuity in both the V_{LD} and V_{RD} ranges between the ECT and CAR results. Thus it looks like the different results are obtained for different orbital levels. As principally a lot can happen in between, I'm puzzled that there isn't at least a comment. Even if for some orbitals the results may not look so nice, a larger panel showing how up and down diagonal results appear in relation to each other would take away the need of wondering what could be suppressed. (Please note though that I comment here on the presentation and I don't put any doubt on the integrity of the data.)

I'm missing moreover a further discussion why the shown results should indeed show ECT and CAR. If a result looks like CAR it doesn't need to be CAR. There is of course further corroboration through the spin resolved measurements, and the filter orientation of the spin split QD levels are shown in the extended data. But here again I found it cumbersome that I had to read through all the text to get all necessary information allowing me to put the full justification together. Why isn't this presented in a compact discussion from the beginning of the paper? Here again I need to trust a

statement long before I'm presented with enough scientific arguments that make me believe the statement, and the scientific evidence is not presented in a compact way but scattered all over text and supplement. I will just add another example, but it isn't my job to rewrite the paper that leaves me the impression of being rather patched together than carefully written with the reader in mind.

For the spin-orbit field B_{SO} I was confused from the beginning and had to jump back and forth in text and supplement to get all information together. First B_{SO} was used and discussed without telling its direction, which I first extracted just from seeing the figure, then there appeared more in the text, and only at the end of the discussion the out of plane angle of 18 degrees was mentioned. Confusion arises because with spin-orbit couping spin itself is no longer a good quantum number but on the one hand its effect is treated as a pure Zeeman field (and added to the external field), and on the other hand its effect is also necessary to produce triplet components. This seems a contradiction. Only through the theory discussion in the supplement (which a reader should indeed consider only for unessential information though) the resolution could be put together with some effort. But what remains still unclear to me is how the authors can use B_{SO} as a Zeeman field at all, since it needs to preserve time reversal symmetry. I know that in carbon nanotubes one can get along with this picture because B_{SO} has opposite signs in different valleys. But it remains unclear to me how that should apply in a semiconductor without resolving in k. Thus here is another piece of physics I'm missing to be able to appreciate the claims of the paper.

Two more things, but not on the presentation. Firstly the number of 71% mentioned in line 111 is 1/sqrt(2) and doesn't refer to the splitting efficiency but to the figure of merit for the Bell test given in Ref. 16. Thus the comparison is wrong. Secondly I'm wondering about the capacitive coupling between the two QDs. The CAR and ECT plots show the results at 45 degrees which suggests no cross-talk of the V_{LD} and V_{RD} gatings on the other dots. I'm missing a comment on that confirming this impression. This comes in addition to a lacking estimate on direct hybridization between the QD. There is of course the superconductor in between, but it isn't too broad to exclude overlap between the QD levels by default. If there was hybridization then one should see current expected to go out through one dot to emerge through the other dot too and, in addition, see anticrossings whenever left and right QD levels become degenerate. I don't perceive this to happen in the data, but the authors need to comment on that because such hybridization scrambles the spin filtering and could thus void the conclusions on triplet detection.

Let me conclude with item (3), the significance of the results. In spite of my comments on the presentation above, this work contains excellent data and a significant advancement in the manipulation and detection of split Cooper pairs. The analysis with just showing the CAR and ECT results comes one analysis step short though. I mentioned above already that I'm lacking a discussion why what looks like CAR is indeed CAR, and for the identification of Cooper pair splitting methods have been deviced that indeed can be used for further validation. The authors mention correctly that Ref. 34 presents related work (although I think the mere mentioning in a footnote lacks a bit of tactfulness) and both works complement each other very well. But Ref. 34 relies on further analysis tools of correlation witness measures (see Phys. Rev. B 96, 064520 (2017), Phys. Rev. Lett. 111, 136806 (2013)) that make the identification of CAR more compelling. Addition of such an analysis would make the results stronger. Furthermore the outlook to creating artificial Kitaev chains will depend on what percentage of the splitting can be converted to a triplet. It seems to me that the

amplitude is still too weak, and a quantitative estimate of what the situation is currently and what still needs to be done would provide a better placement of the results in context than the rather unspecific current statements, which as the most concrete hint tell that the wires are still too short.

Referee #2 (Remarks to the Author):

The work by Wang, Dvir, Mazur et al., investigates Cooper pair splitting in hybrid InAs nanowires. The authors have created quantum dots next to their hybridized nanowire in order to suppress Andreev reflection and they report systematic studies of elastic co-tunneling (ECT) and crossed Andreev reflection (CAR). By properly tuning the bias voltages of the left and right lead and the electrochemical potentials in the two quantum dots they can distinguish between the two processes. In a next step the effect of spin blockade at zero magnetic field on both ECT and CAR is investigated. As a last point they investigate how CAR and ECT behave in the presence of a magnetic field. They report data, which they interpret as the result of triplet Cooper pair pairing.
The submitted manuscript will be of high interest to the broad field of hybrid semiconductor-superconductor devices beyond Majorana physics. I find the work very interesting and very well presented. The data is clear and the analysis is performed in a careful manner. I believe that the work is suitable for Nature.
Before I would like to ask the authors to comment on the following points:
1. Line 33: Can the authors comment on how they tune their device in order to reduce 2-e AR to ~5%? I guess it is important to have a large charging energy but still allow CAR to take place?
2. How important is the length of the proximitized part of the NW for the CET processes? Is there a proximitized quantum dot formed in that part of the device?
3. What is the role of VPG in the experiment? Why is it tuned in one device to zero and in the other there is a positive voltage?
4. Line 73: Can the authors show data of how the diamonds evolve upon tuning the barrier close to the superconductor? How do they tune their devices to block 2-e AR but still allow CAR?
5. Can the authors comment on what is the current flowing through their device (for example in Fig2.) which is not along a diagonal? In 2d, on the top left of the blue line and in 2e on the top right above the red line?
6. In Fig.2b and c the authors state that grey areas indicate the transport window. Does it represent the bias window shown by the white line in Fig. ED9?
7. Line 110: The authors reach an impressive Cooper pair splitting efficiency. Can they comment what they do differently compared to the previous works existing in the literature? What is the "secret" to achieve this?
8. Fig3a: The (0,0) configuration is a (2N,2N) configuration? If yes do the authors have an estimation of how many charges their quantum dots host? How did the authors tune their device to this regime? Do they observe the same physics also for different charge configurations/regimes?
9. Line 175: The authors state that for B//Bso the spin is a good quantum number. I am a bit puzzled by that statement. Once there is spin-orbit one has a pseudospin, i.e. not a real spin any more. Do the authors base their statement on the fact that they did not see an avoided crossing in the excitation spectra in ED3 when B//Bso? In addition and again about Fig. ED3: at what VLD and VRD

values have the measurements been taken? Are the voltages used to define the QDs similar to those used for figures 3 and 4?
10. I am surprised by the statement that the hybrid section has significantly larger SOC. Also by the fact that their sample with and without Pt shows qualitatively the same behavior. The authors state in their recent arXiv (Spin-mixing enhanced proximity effect in aluminum-based superconductor-semiconductor hybrids) “These heavy atoms introduce spin-orbit scattering, which prevents spins from being polarized by an external magnetic field”. So should the two samples (device A and device B) not behave differently?
11. In order to induce triplet correlations in the two QD, does the spin quantization axis in the two dots need to be the same?
12. It is not clear to me how in the supplementary information they conclude that the expected triplet current at phi=0,180o should be no more than 6%. Can they please explain this further?
13. Fig ED1: Can the authors explain how they tune up the devices?
14. Fig ED7d. At which VLD has this plot been taken?
15. The authors have been focusing on 4 configurations in their current versus VRD and VLD plots. Do the devices show similar behavior for different gate voltage configurations? Is the CAR process indicating triplet superconductivity observed for a wide range of gate voltages?

Referee #3 (Remarks to the Author):

The manuscript by Wang et al. reports on Cooper pair splitting experiments with spin sensitivity. Two electrons of a Cooper pair have opposite spin (for ordinary s-wave superconductors) and by splitting them to two quantum dots have been theoretically proposed to yield spatially separated entangled electrons since long. While the manuscript doesn't report on entanglement, it achieves an important milestone by demonstrating a high fidelity spin selectivity and spin selection rules that apply for the splitting process. In particular the spin blockade results of Figs. 3 and 4 in the context of split pairs are conceptually new and the results presented around them are clear and sound. I find this content to fit well to the scope of the journal.

In addition to the above results, the authors use theory arguments to claim that the superconductor-semiconductor hybrid section that they use would have triplet pairing in it. Their argumentation is based essentially on the fact that the length of the quantum dots is shorter that the superconductor-semiconductor hybrid section and hence the spin orbit coupling (SOC) in the hybrid section would to be stronger than in the quantum dot. While this would true, the argumentation does not take into account that during the tunneling process, the electron moves much longer distance than the quantum dot size, and as this distance is comparable or larger than the hybrid segment, the SOC during the tunneling event potentially have the biggest impact to flip the spin. Also, as the authors correctly point out, the fact that the electronic states in the hybrid segment need to be at least partially in the aluminium segment, the SOC in the hybrid part is suppressed from the strong SOC of the bare InSb. Also, the manuscript does not present any direct experimental evidence about the triplet pairing but rather probes the spin directions outside of the superconductor. For a compelling evidence of the triplet p-wave pairing, one should present results about the formation of the Kitaeev chain, not done in this work. Because of these reasons, I find the claims about the triplet pairing

non-conclusive and speculative and they should be removed from the manuscript before publishing. In particular, I find the claim "Our results demonstrate controllable detection of singlet and triplet pairing" in the abstract misleading. As stated above, I find the manuscript content excellent but such overstatements about the p-wave pairing should be removed.

More minor comments/questions on the manuscript content:

1) Direction of the B_SO: I don't see any arguments why the direction of the B_SO field should be along the y-axis of Fig. 4a. What is the argumentation for this particular direction? This would be worth clarifying. I presume the nanowire may land at an arbitrary angle to the substrate and hence the B_SO can be at any angle in the y-z plane. It is also claimed that the device B has a 30 degree angle between the substrate surface and B_SO. How was this estimate obtained? It would be also good if the authors could comment on why the Dresselhaus contribution is neglected in the analysis.

2) Caption of Fig 4, line 156: "applied B direction defined as the spin quantization axis." and main text line 170: "We define the applied B direction as our spin direction axis" are somewhat sloppy formulations. It's rather that you apply a magnetic field to a specific direction and as a result, the magnetic field sets the spin quantization axis. It's a matter of what field you apply in the experiment, not a matter of what definition you take for the applied magnetic field. I suggest to reformulate these sentences accordingly.

3) Fig. ED2 and text around line 414: the background current I_BG seems to be taken in Coulomb blockaded regime where charging energy cost suppresses the transport. A more truthful estimate of the background level could be taken inside the transport windows at the anti-diagonal areas. Inside the transport window, sequential single-electron tunneling is possible without a charging energy cost, as well as possibly other competing processes, which may lead to lower values of the splitting estimates than what is reported now.

4) Fig. ED6 panels e-p: These would be easier to read if they would be arranged into a 2x2 matrices similarly as the data in Figs. 3c and Fig. 4c and f. With the current organization of the panels, it is very laborious to assess if the data follows the correct structure in line with what is presented in the main manuscript figures.

5) Fig. ED8: Magnetic field magnitude and voltage settings are not given. When applying perpendicular field as done here, I would be expecting the superconductivity to become suppressed and influence on the analysis. Also, what relevance does the 18.3 degree angle have? I don't see what it the point of this fitted value.

6) References: Since this study is about new spin blockade effect, it would be good to include in the discussion the pioneering work of K. Ono et al, Science 297, 1313 (2002) about the Pauli spin blockade. Also the more recent works A. Hofmann et al, Phys. Rev. Lett. 119, 176807 (2017) and Ji-Yin Wang et al. Nano Letters 18, 4741 (2018) that discusses the SOC and its anisotropy in spin blockade would probably be useful references for the B_SO argumentation and discussion.

Author Rebuttals to Initial Comments:

We wish to thank all three referees for their careful reading of our report and for the constructive comments allowing further improvement in the quality of manuscript. Below we list the major changes we implemented in our paper to answer these comments:

1. We add a discussion on the results shown in Fig 2 to justify their interpretation as resulting from CAR and ECT.
2. We improve the introduction of B_{SO} and clarify that the CAR and ECT measurements are used to determine B_{SO} direction instead of it being known a priori. We also mention the B_{SO} direction thus measured can assume different values depending on the device and gate settings.
3. We have added a spin-correlation analysis of the results shown in Fig 4. The analysis is explained in a new Fig ED8. As pointed out by the referees, the results shown in the previous Fig ED8 did not contribute to make the explanation clear and has been removed.
4. We describe in detail in the Methods section the tuning procedure of our device. We further explain the reasons that our experimental platform gives superior results compared to prior state-of-the-art.
5. We remove from the paper definitive claims of observing triplet paring in the proximitized superconductor. (In fact, we did not intend to imply we proved p-wave superconductivity in the hybrid, although we acknowledge the previous version was ill-phrased in this respect. The physics reported here is insensitive to whether the hybrid segment is topological or trivial.) We now clarify the main claim is that we observe a triplet component in the pairing between QDs and discuss the possible mechanisms for this. We point to our other work (arXiv:2206.08045), where the same device was tuned to the strong coupling limit, as another supporting evidence for our claim of triplet pairing between the QDs.

A point-by-point response to all comments made by the referees is given below.

Referee #1 (Remarks to the Author):

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Item (1) is of high standard. The obtained linewidths, the stability diagrams, seeming lack of cross-talk between the dots show all an excellent sample quality and experimental access. The main results presented are supported by several well chosen and indeed required extended data plots. There isn't much I can add, this is state-of-the-art.

We thank the referee for the critical review of our paper and the positive assessment of the quality of the results.

On the other hand, items (2) and (3) have shortcomings. The analysis and presentation, item (2), have made only an incomplete transition from what is clear to the authors to what the reader will need to know and when. I leave the issue open that the text would be inaccessible to anyone not fully familiar with superconducting nanostructures, and I start from considering that e.g. the provided description of CAR, ECT etc. is sufficient.

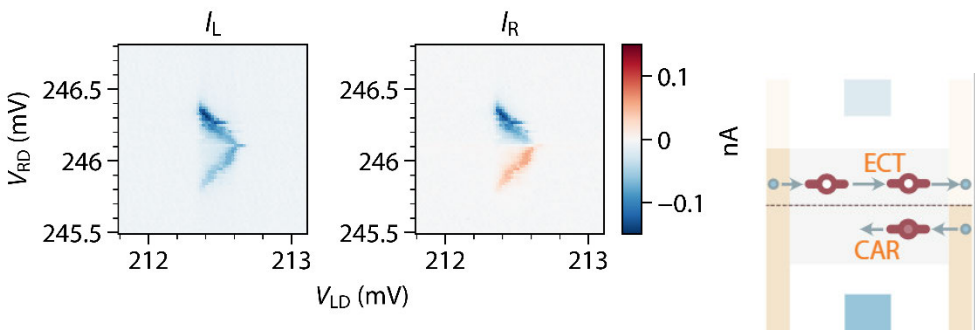
We agree with the referee that we assume the basics of knowledge in superconducting nanostructures and refer to Prada, et al., *Nat Rev Phys* **2,** 575–594 (2020) for background information. Our starting point is indeed a brief introduction to CAR and ECT.

Yet I find quite a number of important statements made in a way that implies more that one should trust the authors rather than allowing the readers to verify the conclusions themselves. Let me start with Fig. 2. From the sketches in panels b and c one can indeed infer the shown signals and their signs on the diagonals of the following panels. But from the sketches one would also assume that a convincing evidence for the cross-over from ECT to CAR should show up from the changing signal when continuously lowering of V_{LD} at fixed V_{RD}. If one looks at the voltage ranges, however, there is a discontinuity in both the V_{LD} and V_{RD} ranges between the ECT and CAR results. Thus it looks like the different results are obtained for different orbital levels. As principally a lot can happen in between, I'm puzzled that there isn't at least a comment. Even if for some orbitals the results may not look so nice, a larger panel showing how up and down diagonal results appear in relation to each other would take away the need of wondering what

could be suppressed. (Please note though that I comment here on the presentation and I don't put any doubt on the integrity of the data.)

Before replying to the specific question made here, we would like to point out that measuring CAR requires two experimental conditions as explained in the manuscript: 1. equal biases on both N leads, and 2. opposite chemical potential of the QDs. ECT requires two opposite conditions: 1. opposite biases on both N leads, and 2. equal chemical potential of the QDs. The measurements shown in Fig. 2 are conducted either with equal biases (to show CAR) or opposite biases (to show ECT). Therefore, tuning V_{LD} while keeping V_{RD} fixed cannot lead to a crossover from a process allowed by the fixed bias polarities to one that is blocked by the bias polarities.

However, when one bias is kept at zero while the other at finite voltage, a crossover from CAR to ECT induced by the QD gate is indeed observable within one charge degeneracy. This can be seen in Fig ED6 (Device B) and the underlying data in Fig 4g. The inset sketch in Fig ED6 panel c illustrates why this is the case (attached below). In short, the unbiased N lead allows both electrons and holes to tunnel in and therefore both electron-electron and electron-hole tunneling can be measured. To give a clear example of this type of measurement, we show a zoom-in to one joint charge degeneracy where one of the biases (V_R in this example) was set to zero. In these settings, a crossover from CAR currents to ECT currents is possible by only changing the QDs chemical potential (V_{RD} in this case) in the vicinity of one charge degeneracy:



CAR and ECT can be distinguished both by examining the sign of the current on the right lead, and by looking at the slope of the signal in the plot.

Having said that, indeed, the measurements presented in Fig. 2 were taken with different QD orbitals than the rest of the paper because they are intended to show the highest-quality data we have for each of these two process. Since the observations highlighted by these datasets and discussed in the relevant text hold for virtually every QD charge degeneracy we measured, we chose the data with the highest resolution and Cooper pair splitting efficiency and explained as such in the caption of Fig ED1. We now move this description to the main text (lines 261-264):

"We note that the data shown in Fig. 2d and Fig. 2e are taken with different gate settings than the rest of the paper and are selected because of high data resolution and Cooper pair splitting efficiency."

Additionally, the ECT measurements were taken at $V_{PG} = 0.18$V and $B_x = 0.1$T. We now mention this explicitly in the caption of Fig 2.

A fully consistent and comprehensive data set as the referee asked, where CAR and ECT are measured for the same QD orbitals, is shown in Fig. 3 (panel c) and 4 (panels c, f). In these datasets, only B and the bias polarities are varied (and all four combinations are presented) while gate voltages are fixed or swept in the same, large range. The full, unprocessed data are shown in Fig ED10. In particular, Fig. 3a also shows such a full gate range in which four QD resonances appear without interrupted axes.

I'm missing moreover a further discussion why the shown results should indeed show ECT and CAR. If a result looks like CAR it doesn't need to be CAR. There is of course further corroboration through the spin resolved measurements, and the filter orientation of the spin split QD levels are shown in the extended data. But here again I found it cumbersome that I had to read through all the text to get all necessary information allowing me to put the full justification together. Why isn't this presented in a compact discussion from the beginning of the paper? Here again I need to trust a statement long before I'm presented with enough

scientific arguments that make me believe the statement, and the scientific evidence is not presented in a compact way but scattered all over text and supplement.

We argue that the results presented in Fig 2. show convincing evidence for CAR and ECT, independently of the later discussion of the spin resolved measurements. To see this, we follow the knowledge in the field that sub-gap transport can take place via one of the few processes:

1. Resonant quasi-particle tunneling into and out of sub-gap bound states
2. Andreev reflection at the interface with the superconductor
3. CAR
4. ECT

We can rule out (1) since spectroscopy shows no sub-gap states exist in the bias range we apply (this can be seen by observing the Coulomb diamonds in Fig ED1 or the energy spectrum in Fig ED9). We can rule out (2) by noticing that tuning one QD on resonance is not enough for current to flow through the system. It is required that both QDs are on resonance, showing that the transport is highly non-local. While these elements above have served as enough proof for the presence of CAR and ECT in previous publications, our results go further: we show that perfectly correlated positive currents occur only when the energies of the QDs are opposite, and perfectly anti-correlated currents occur only when the energies of the QDs are aligned. The high degree of side-to-side correlation can be seen from the fact that even gate jitters show up identically in both I_L and I_R (e.g., V_{RD}=210.9mV in Fig 2d and V_{RD}=200.3mV in Fig 2e). We are not aware of any other phenomena except CAR and ECT that can give rise to such observations. Additionally, previous reports also considered higher-order processes that involve multiple CAR/ECT steps and multiple intermediate QD states. These are also unlikely to be significant in our measurements since they generally lead to deviation from the ideal $|I_L| = |I_R|$ relation (Schindele, et al., PRL 109, 157002 (2012)).

To clarify this point we add to the main text the following (lines 235 – 246):

"Several features in these data allow us to attribute the origin of these sub-gap currents to CAR and ECT instead of competing transport processes. The non-local origin of the measured currents, expressed by the (anti-)symmetric energy requirement on both QDs and current correlation, rules out local Andreev reflection. The bias and QD energies being kept lower than any sub-gap bound state excludes resonant tunneling into and out of them. The only mechanisms known to us that can explain these observations are CAR and ECT [31]."

I will just add another example, but it isn't my job to rewrite the paper that leaves me the impression of being rather patched together than carefully written with the reader in mind.

For the spin-orbit field B_{SO} I was confused from the beginning and had to jump back and forth in text and supplement to get all information together. First B_{SO} was used and discussed without telling its direction, which I first extracted just from seeing the figure, then there appeared more in the text, and only at the end of the discussion the out of plane angle of 18 degrees was mentioned.

We acknowledge the comment made by the referee and agree that the description of B_{SO} can be confusing. The source of the confusion lies in the fact that the gates settings in which ED8 is taken are different than the gates settings with which the data in Fig 3 and Fig 4 were taken. This is important, as the precise values of the applied voltages on the gates affect the direction of B_{SO} as we are currently measuring in on-going experiments.

In fact, the determination of the B_{SO} direction is an open question in hybrid systems. We assert that our experiment has an ability to shed extra light on this question – both equal-spin CAR and opposite-spin ECT require the breaking of spin conservation and are thus suppressed when B // B_{SO}. We propose here to use the observed spin conservation as a method to determine the direction of B_{SO}.

In the case of the data of Fig. 3 and Fig. 4, we only measured CAR and ECT while rotating the magnetic field in-plane (the *x-y* plane) and cannot determine the out-of-plane component of B_{SO}. However, since the suppression of the non-spin conserving processes is rather complete when B // y, we reasonably conclude that the out-of-plane component, if any, is small.

To avoid this confusion, we modify our main text by explaining the above (404 – 472):

"We first apply B = By = 100 mT, in the plane of the substrate and perpendicular to the nanowire. The four panels in Fig. 4c present I_{corr} measured at four bias polarity combinations, selecting either CAR or ECT conditions. The upper left panel also shows the lowest-energy spin combinations. I_{corr} vanishes for ↑↑ and ↓↓ with CAR biases –– and ++, and for ↑↓ and ↓↑ with ECT biases +– and –+. The observation of spin conservation suggests spin is a good quantum number. Thus, any spin-orbit field in the InSb nanowire, B_{SO} (including both possible Rashba and Dresselhaus SOC), must be parallel, or nearly parallel, to By. In this case, the superconducting pairing between the QDs is singlet. We note that the B_{SO} direction as measured by suppression of equal-spin CAR/opposite-spin ECT depends on gate settings and the device used (e.g., Fig. ED6). We have measured directions within 20° of being perpendicular to the nanowire axis but its angle with the substrate plane can range from 0 to 60°. This observation is consistent with the expectation of BSO being perpendicular to the nanowire axis for both Rashba and Dresselhaus SOC [38–40]."

> Confusion arises because with spin-orbit couping spin itself is no longer a good quantum number but on the one hand its effect is treated as a pure Zeeman field (and added to the external field), and on the other hand its effect is also necessary to produce triplet components. This seems a contradiction. Only through the theory discussion in the supplement (which a reader should indeed consider only for unessential information though) the resolution could be put together with some effort. But what remains still unclear to me is how the authors can use B_{SO} as a Zeeman field at all, since it needs to preserve time reversal symmetry. I know that in carbon nanotubes one can get along with this picture because B_{SO} has opposite signs in different valleys. But it remains unclear to me how that should apply in a semiconductor without resolving in k. Thus here is another piece of physics I'm missing to be able to appreciate the claims of the paper.

The analysis of the SOC in our system is based upon past works showing how SOC in double quantum dots leads to spin-flip tunneling (Nadj-Perge et al., Nature 468, 1084 (2010), Hofmann et al, Phys. Rev. Lett. 119, 176807 (2017), Wang et al. Nano Letters 18, 4741 (2018)). As the referee points out correctly, the spin-orbit field does not break time reversal symmetry and we do not treat it as a Zeeman field at all. The spin orbit operator has a term in the form $B_{SO}\sigma_y = \alpha\sigma_y p_x$ where σ_y is the Pauli matrix acting on the spin degree of freedom and p_x is the momentum operator associated with the tunneling between the two QDs. This is indeed slightly different from a bulk system with a well-defined crystal momentum *k*, but the physics remains that a special axis exists which is perpendicular to both the inversion-breaking electric field and electron momentum. When the magnetic field is applied parallel to this direction, i.e., B // B_{SO}, spin is conserved in the tunneling process. When the magnetic field is applied perpendicular to this direction, there is a finite probability for a spin flip. A classical, intuitive analogy of the spin-flip tunneling is often adopted in literature, namely spin precession along the B_{SO} direction (Stano et al., Phys Rev B 72, 155410 (2005)). To clarify the possible confusion, we cite the relevant literature reporting anisotropic spin-flip tunneling in double QDs and explain that the spin-precession picture is a classical analogy instead of a precise quantum description.

> *Two more things, but not on the presentation. Firstly the number of 71% mentioned in line 111 is 1/sqrt(2) and doesn't refer to the splitting efficiency but to the figure of merit for the Bell test given in Ref. 16. Thus the comparison is wrong.*

We thank the referee for pointing out this confusion. Ref. 16 presents three terms, all related to the ratio of current (or conductance) arising from CAR relative to the total current including competing processes. The first term, the *CPS visibility*, (CPS stands for Cooper pair splitter) is defined for each lead as the ratio of the excess conductance due to CAR over the total conductance for a given lead. This term is calculated in ED2 (using current rather than conductance) and we arrive at values of 91.3% and 98% for the left and right, respectively. We (mis)labeled this term as *efficiency* while it has been defined in literature as *visibility*. The second term introduced in Ref. 16 is the *CPS efficiency* which is the ratio between CAR current and the total current in both leads. This is equivalent to the average of the two visibilities above and is around 95% rather than the 90% quoted in the previous version of our manuscript. Finally, Ref. 16 introduces a figure of merit related to current cross correlations on both leads, which is the product of the *visibilities* of both leads. This figure of merit is required to be larger than 1/sqrt{2} for a Cooper pair splitter to be able to observe a potential violation of Bell's inequality. This figure of merit is what we report in our main text, not *total efficiency* as originally labeled. We emphasize that the value we obtain for the figure of merit, nearly 90%, satisfies the mentioned readout requirement to observe a violation of Bell's inequality.

We now correct the wrong labeling of these terms in lines 269-276 of the main text and in the caption of Fig ED2.

"In Fig. ED2 we extract from this measurement the Cooper pair splitting visibilities of 91% and 98% for the left and right QDs, respectively. Their product of 90%, for the first time realized, exceeds the minimum value of 71% required for a Bell test [16]."

> Secondly I'm wondering about the capacitive coupling between the two QDs. The CAR and ECT plots show the results at 45 degrees which suggests no cross-talk of the V_{LD} and V_{RD} gatings on the other dots. I'm missing a comment on that confirming this impression.

The referee points out correctly that the observed diagonals in the CAR and ECT plots indicates no cross coupling between the gates controlling the two QDs. This is also evident in Fig. 3 where the multiple resonances of the double dot system appear on a rectangular grid, and also in Fig ED10 where the local leakage currents on the right lead appears as horizontal lines. This is a reasonable result as the two gates are separated by three other metallic gates and a superconducting film in between that screen stray electric fields.

We modify our text by noting this explicitly (lines 158-161)

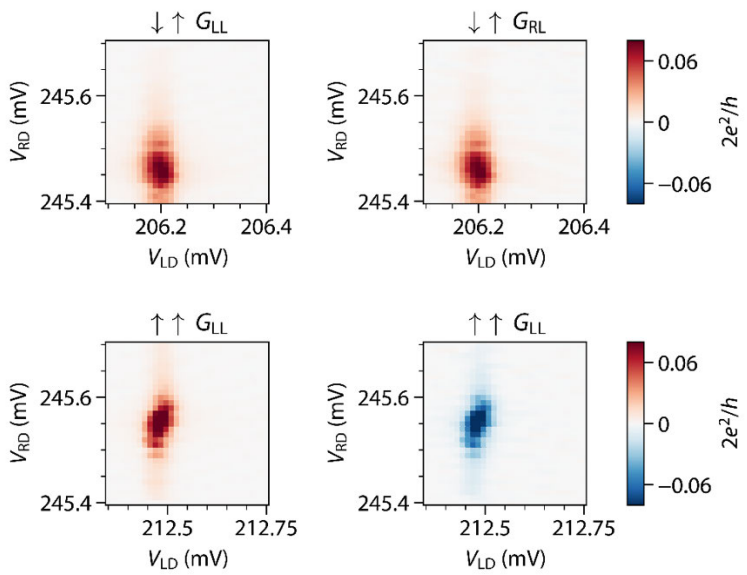
"We note that screening due to the presence of multiple metallic gates and a superconducting film in between diminishes cross-coupling between V_{LD} and V_{RD}."

> This comes in addition to a lacking estimate on direct hybridization between the QD. There is of course the superconductor in between, but it isn't too broad to exclude overlap between the QD levels by default. If there was hybridization then one should see current expected to go out through one dot to emerge through the other dot too and, in addition, see anticrossings whenever left and right QD levels become degenerate. I don't perceive this to happen in the data, but the authors need to comment on that because such hybridization scrambles the spin filtering and could thus void the conclusions on triplet detection.

As the referee points out, the QDs can, in principle, hybridize due to CAR and ECT interactions. In a related work, we tune Device A to reach the strong coupling limit where QDs indeed strongly hybridize and anticrossings show up clearly in the data (arXiv:2206.08045). This is done by lowering the relevant tunnel barriers to increase the QD-QD coupling strength. In this work, however, it is desirable to stay in the weak-coupling regime so that the QD eigenstates are unperturbed and can serve as good charge/spin filters.

Hybridization anticrossings are observable when the coupling strength between the QDs exceeds both $k_B T$ and the linewidth of the QDs. In the Supplementary Material, we present an estimation of the QD-QD and QD-lead couplings using Eq. 1. The extracted couplings were 9 and 4 μeV for ECT and CAR in Fig. 2, respectively. These couplings are of the order of $k_B T$, but much smaller than the QDs' linewidth (30 – 70 μeV), and should have a correspondingly weak effect on the observations. Similarly, for the spin-resolved data presented in Fig 3 and Fig 4, the coupling strengths are approximately 7 μeV, confirming that hybridization between the QDs are lower than the broadening energy scales.

To answer the referees question directly, we also show the absence of anticrossings in the (V_{LD}, V_{RD}) plots. Such anticrossings can, in principle, be observed when measuring the conductance while applying zero DC bias and scanning the chemical potential of both QDs. We plot below examples of such measurements using the same gates settings as those of Fig 3 and 4, showing no anticrossing.



Let me conclude with item (3), the significance of the results. In spite of my comments on the presentation above, this work contains excellent data and a significant advancement in the manipulation and detection of split Cooper pairs.

We appreciate the referee's positive feedback on the significance of our work.

The analysis with just showing the CAR and ECT results comes one analysis step short though. I mentioned above already that I'm lacking a discussion why what looks like CAR is indeed CAR, and for the identification of Cooper pair splitting methods have been deviced that indeed can be used for further validation. The authors mention correctly that Ref. 34 presents related work (although I think the mere mentioning in a footnote lacks a bit of tactfulness) and both works complement each other very well. But Ref. 34 relies on further analysis tools of correlation witness measures (see Phys. Rev. B 96, 064520 (2017), Phys. Rev. Lett. 111, 136806 (2013)) that make the identification of CAR more compelling. Addition of such an analysis would make the results stronger.

We accept the suggestion made by the referee and add the spin correlation to our discussion of the results of Fig. 4 (new Fig ED8). We add the following text (lines 472-481)

"To quantify the observation that CAR is anti-correlated with the spin alignment of the QDs, we perform a spin correlation analysis similar to that in Ref [41], which analogously reports reduced CAR amplitudes when QD spins are parallel compared to anti-parallel. The results, presented in Fig. ED8, show two QD spins are anti-correlated by a factor of –0.86 for CAR signals when pairing is singlet, the highest reported to date."

The calculated CAR spin correlation reaches a minimum of –0.86 when B // B_{SO} and spin pairing is purely singlet. This is among the first reports of strongly anti-correlated spin measurements of split Cooper pairs together with Bordoloi et al, who reported maximum spin anti-correlation of –0.37. The currently reported figure is likely limited by our method of estimating CAR and ECT strengths (simply taking the maximum/minimum of measured current) since they are sensitive to measurement noise.

When B is not parallel to B_{SO}, the spin correlations begin to deviate further from –1. Such anisotropy is expected and, as explained in the main text, a testimony of spin-orbit-induced spin triplet pairing. The deviation from a simple spin anti-correlation due to spin-orbit interaction lies at the foundation of future entanglement witness experiments as proposed in PRL 111, 136806 (2013).

Furthermore the outlook to creating artificial Kitaev chains will depend on what percentage of the splitting can be converted to a triplet. It seems to me that the amplitude is still too weak, and a quantitative estimate

of what the situation is currently and what still needs to be done would provide a better placement of the results in context than the rather unspecific current statements, which as the most concrete hint tell that the wires are still too short.

In our related work (arXiv:2206.08045, currently under review), we tune the same device reported here to the Kitaev chain regime. Indeed, the creation of a Kitaev chain requires equal amplitude of ECT and CAR. The data presented in Fig. 4 of the main text shows that the maximal equal spin CAR current is smaller than the equal spin ECT current. Choosing the opposite spin configuration results in the opposite problem where the ECT current is smaller than the CAR current. However, in Fig ED6 (Device B), we already observe the ECT and CAR currents can be equal for some directions of the applied field in one spin configuration (see Fig. ED6b).

Another challenge on the path towards a Kitaev chain is to increase the coupling strength between the QDs. The Kitaev chain is formed when adjacent QDs hybridize by CAR and ECT interactions which requires the coupling strength to be at least larger than $k_B T$ and QD linewidth.

We accept the referee's suggestion to discuss the requirements of a Kitaev chain in a more concrete manner and modify the conclusion segment as follows (lines 6321- 640)
"In conclusion, we have measured CAR and ECT in an N-QD-S-QD-N device with and without spin filtering. For well-defined, specific settings consistent with our expectations, we observe Cooper pair splitting for equal spin states in the QD probes. These observations are consistent with the presence of a triplet component in the superconducting pairing in the proximitized nanowires, which is one of the building blocks for a topological superconducting phase [48, 49]. More generally, our results show that the combination of superconductivity and SOC can generate triplet CAR between spin-polarized QDs, paving the road to an artificial Kitaev chain [7–9]. The realization of a Kitaev chain further requires increasing the coupling strength between QDs to allow the formation of a hybridized, extended state. This is confirmed in a parallel work where the QDs are driven to the strong coupling regime [50]."

Referee #2 (Remarks to the Author):

The work by Wang, Dvir, Mazur et al., investigates Cooper pair splitting in hybrid InAs nanowires. The authors have created quantum dots next to their hybridized nanowire in order to suppress Andreev reflection and they report systematic studies of elastic co-tunneling (ECT) and crossed Andreev reflection (CAR). By properly tuning the bias voltages of the left and right lead and the electrochemical potentials in the two quantum dots they can distinguish between the two processes. In a next step the effect of spin blockade at zero magnetic field on both ECT and CAR is investigated. As a last point they investigate how CAR and ECT behave in the presence of a magnetic field. They report data, which they interpret as the result of triplet Cooper pair pairing.
The submitted manuscript will be of high interest to the broad field of hybrid semiconductor-superconductor devices beyond Majorana physics. I find the work very interesting and very well presented. The data is clear and the analysis is performed in a careful manner. I believe that the work is suitable for Nature.

We thank the referee for the positive evaluation of our work.

Before I would like to ask the authors to comment on the following points:
1. Line 33: Can the authors comment on how they tune their device in order to reduce 2-e AR to ~5%? I guess it is important to have a large charging energy but still allow CAR to take place?

The referee is indeed correct in his assumptions that the interplay between the charging energies of the QDs and their coupling to the hybrid segment determines the efficiency of CAR/ECT (both being 1*e* processes) vs local Andreev reflection (a 2*e* process). The approach we take is to fabricate our devices such that the charging energies of the confined dots are on the order of 2 meV.

We add a detailed description of our QD tuning protocol in Methods (lines 697-730):

"Device tune-up

The tuning of our device, in particular the QDs, is done as follows. First, we form a single barrier between N and S by applying a low voltage on the gate closest to S on each side. We then perform local and non-local tunnel spectroscopy of the hybrid segment and locate a V_{PG} range in which a hard gap is observed at low energies and extended Andreev bound states are observed at high energy (see Fig. ED9). Having located a desired value of V_{PG}, we form a second barrier in each junction by applying a lower voltage on the gates closest to the N leads. The confined region between the two barriers thus becomes a QD. We characterize the QDs by measuring its current above the superconducting gap, applying $|V_L|$, $|V_R| > \Delta/e$ as a function of V_{LD}, V_{RD} and applied magnetic field (see Fig. ED1c,d). We look for a pair of resonances that correspond to the filling of a single non-degenerate orbital. This is indicated by two resonances separated by only the charging energy at zero field and their linear Zeeman splitting when B > 0. We finally measure CAR and ECT between the two QDs (as discussed in Fig. 2). We optimize the measurement by controlling the gates separating the QDs from S to balance low local Andreev current (lowering gate voltage) with high signal-to-noise ratio (raising gate voltage). Having reached a reasonable balance, we characterize again the QDs (Fig. ED1).

2. How important is the length of the proximitized part of the NW for the CET processes? Is there a proximitized quantum dot formed in that part of the device?
3. What is the role of VPG in the experiment? Why is it tuned in one device to zero and in the other there is a positive voltage?

Questions 2 and 3 of the referee are both related to the properties of the hybrid segment and their influence over the processes of CAR and ECT. The very short hybrid segment forms Andreev bound states – equivalent to a proximitized QD as termed by the referee (however, with vanishing charging energy due to strong coupling to the grounded superconductor). Fig ED9 shows the spectrum of such ABS evolving under a magnetic field. We believe that the CAR and ECT processes we report on in this work arise from virtual excitations of the ABS. The complete description of this physical model and its underlying hopping processes is presented in a parallel theoretical paper (arXiv:2203.00107). The role of V_{PG} in this experiment is to tune the energy and charge composition of these ABSs. Indeed, applying very negative V_{PG} drives the ABS energy up towards the gap edge and quenches CAR and ECT. In

Device A, the ABS was near its energy minimum at $V_{PG} = 0$. In the case of Device B, positive V_{PG} was required to pull the ABS energy down to below the superconducting gap.

These questions are highly relevant and a comprehensive manuscript of the ABS's effect on CAR and ECT is under preparation. Below we show one possible figure from that future publication. Panels b and c show tunnel-spectroscopy results of the hybrid segment while sweeping V_{PG}. We observe a hard superconducting gap and some sub-gap resonances, i.e., various ABSs, appearing at certain values of V_{PG}. Panel e shows the amount of CAR and ECT currents when we tune up QDs at corresponding V_{PG} values. They are clearly correlated with the appearance of the ABSs, especially at V_{PG}=35mV and 110mV. These results were taken during a different cool down of Device A, hence a small shift in the energy of the ABSs with respect to the results appearing in this work.

[redacted]

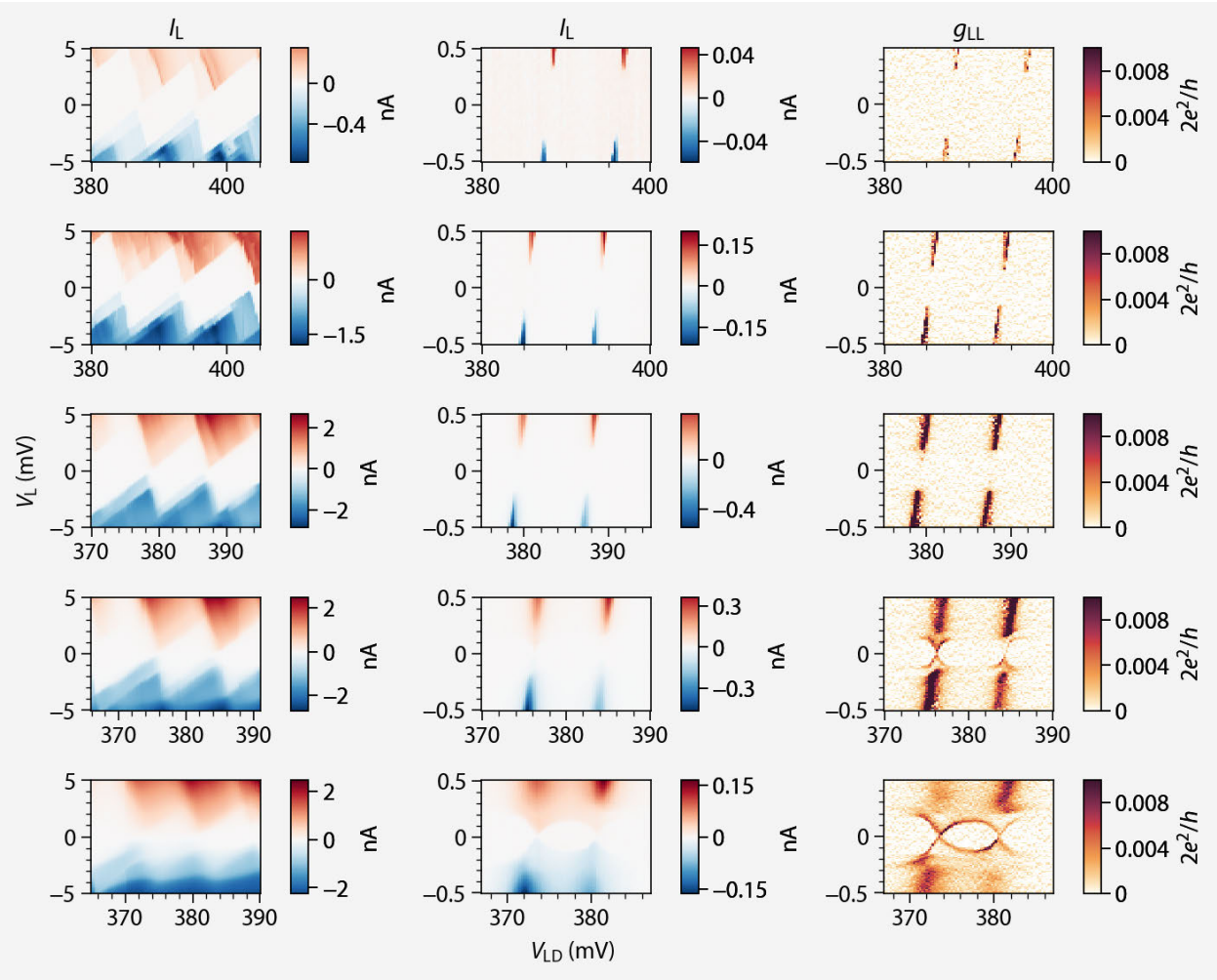
We believe that the length of the hybrid segment affects the amplitude of the QD-QD couplings and the amount of precession induced by SOC in the CAR/ECT processes. While we do not have conclusive evidence for the above, we note that in general the magnitude of currents observed in Device B (350 nm long) were smaller than those observed in Device A (200 nm long). Furthermore, we could tune Device A to the strong coupling regime where both QDs hybridize with each other while we could not do the same with Device B. We further note that the amplitude of equal-spin CAR in Device B can be tuned to be well above the amplitude of equal-spin ECT (not presented in the current manuscript), showing a large amount of spin precession. We stress that these observations depend heavily on the precise microscopic details and we currently cannot derive strong conclusions from them.

> 4. Line 73: Can the authors show data of how the diamonds evolve upon tuning the barrier close to the superconductor? How do they tune their devices to block 2-e AR but still allow CAR?

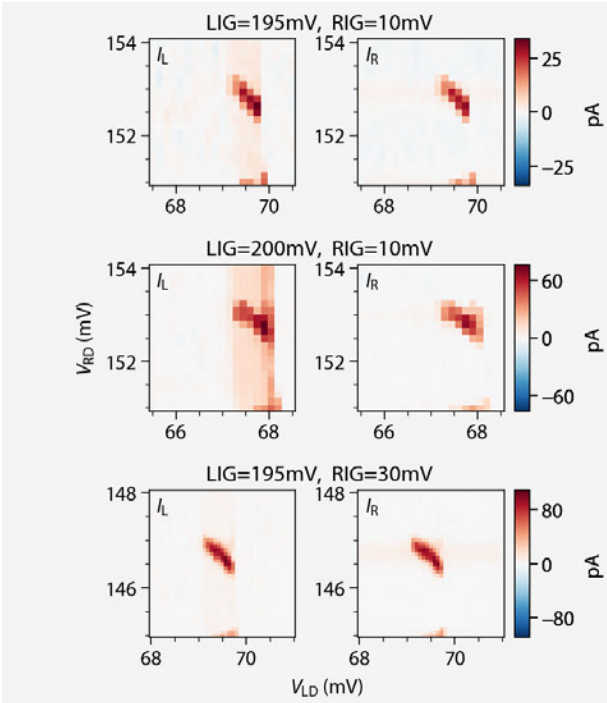
We did not perform the diamond vs inner gate sweeps on the original devices A and B, so we measured this on another similar device currently in the fridge instead. We believe the behavior shown here to be generic. Below we show three plots in a row for each left inner gate (the gate separating the left QD from the superconductor) value: a large-range diamond sweep of the left N-QD-S current, a zoom in to the superconducting gap energies and a numerical derivative of the zoom-in measurement showing conductance. The inner gate voltages from the top row to the bottom row are 10, 30, 50, 60, 70mV.

When the barrier is high (low voltage), only above-gap quasiparticle currents are measurable since these are 1*e* processes not suppressed by charging energy. We note that CAR and ECT are also 1*e* in each junction since no double-occupation on a given QD is required. This is the limit where 2*e* sub-gap current is most strongly suppressed

relative to 1*e*, but the overall signal is also extremely weak. As the barrier gradually opens up, sub-gap currents begin to emerge near charge degeneracies while the overall signal becomes stronger. In the strong QD-S coupling limit (undesirable in this work), sub-gap bound states form and QD levels become proximitized Andreev/YSR levels with typical eye-shaped singlet-doublet transitions.



When tuning-up the devices, we attempt to maximize the amount of CAR and ECT currents while keeping the local Andreev reflection process minimal. This is typically found in the weak-to-intermediate coupling regime. Below we show an example from device A how we tune our devices using the two inner gates next to the left and the right QDs (LIG and RIG, respectively):



In this example, it is clear that applying too positive a gate voltage on the left barrier leads to local Andreev reflection reaching an amplitude similar to that of CAR, which is undesirable. Increasing the voltage on the right barrier, however, although also slightly enhancing the local Andreev reflection, allows for a much greater CAR current, improving the overall CAR signal.

5. Can the authors comment on what is the current flowing through their device (for example in Fig2.) which is not along a diagonal? In 2d, on the top left of the blue line and in 2e on the top right above the red line?

The currents shown in Fig. 2 flowing above the diagonal lines arise from inelastic CAR and ECT processes involving energy relaxation. In the case of Fig. 2d, we show ECT when electrons flow from the left QD to the right QD. While this process is maximized when both QD are resonant with each other, if the left QD has higher chemical potential than the right QD (lower applied voltage), electrons can still tunnel between them while losing the excess energy to the environment. A similar process takes place for CAR in panel e. We point the referee for Fig ED6, where this phenomenon is much more prominent.

We comment on it in the main text (lines 253-261):

"We further notice that finite ECT and CAR currents can be observed when both QDs are within the transport window but not on the diagonal lines dictating energy conservation. Since they appear only on one side of the (anti-)diagonal line corresponding to down-hill energy relaxation, these currents result from inelastic processes involving spontaneous emission and are thus non-coherent."

6. In Fig.2b and c the authors state that grey areas indicate the transport window. Does it represent the bias window shown by the white line in Fig. ED9?

Indeed, the white bar in Fig ED9 shows the value of the bias defining the transport window in the measurements shown in Fig. 4. This is mentioned in figure caption: *"White line indicates the bias range in which the experiments at finite B field were performed".*

7. Line 110: The authors reach an impressive Cooper pair splitting efficiency. Can they comment what they do differently compared to the previous works existing in the literature? What is the "secret" to achieve this?

We thank the referee for acknowledging our achievement. Our setup is different from previous works in the field of Cooper pair splitting in two essential details:

1. Having 7 independent gates to control the setup – three for each QD and one for the hybrid – is essential. The ability to tune the chemical potential of the QD independently from the QD-lead coupling allows us to suppress local AR while maintaining finite CAR and ECT amplitudes. Almost all other works in this field rely on a single gate for each QD, making the tuning of the devices challenging. One-gate QDs often also suffer from large QD energy broadening (~100uV in some previous literature) and cannot achieve the energy resolution we have with these devices.
2. Having a high-quality semiconductor-superconductor interface, giving a hard gap in the density of states along with discrete, finite-energy Andreev bound states. In the case of a soft gap, it is impossible to quench single quasiparticle tunneling without quenching CAR and ECT. Therefore, the presence of a soft gap in the density of states prevents one from reaching high efficiency.

An additional requirement for the spin-resolved part of experiment is to have a grounded thin superconducting contact. All of these technological foundations were developed in our group recently and are described in Heedt, et al., Nat Commun 12, 4914 (2021).

As a final note, we believe that the conditions described above are sufficient to reproduce our experiment, as we are aware of another group, working with a different material platform, that has followed these guidelines and reproduced our results.

We address these points in main text (lines 274-284) :

”**The high efficiency of Cooper pair splitting reported in this work compared to previous reports relies on having a hard superconducting gap in the proximitized segment and on having multiple gates for each QD, allowing control of the chemical potential of QDs independently from QD-lead couplings. Both requirements are enabled by recent advancements in the fabrication technique [33].**”

8. Fig3a: The (0,0) configuration is a (2N,2N) configuration? If yes do the authors have an estimation of how many charges their quantum dots host? How did the authors tune their device to this regime? Do they observe the same physics also for different charge configurations/regimes?

Indeed (0,0) is counted from a reference charge occupation that is even, as noted in line 132-133 of the previous version. We estimate this reference occupation is on the order of ~10 based on Coulomb peak counting and the characteristic length indicated by level spacing. This few-electron regime is easily within reach for InSb nanowires when all gate voltages are near zero.

We observe the same physics for many different QD orbitals/occupations in the few-electron regime (two more examples in reply to question 15), but did not explore the many-electron regime where bipolar spin filtering is not available. The key to obtaining the reported results is to ensure the QD orbitals are well spin-polarized along the applied field direction. This requires large level spacing, two-fold degenerate orbitals at B=0 that linearly disperse with B in opposite directions, staying far away from crossings of opposite spin states (e.g., when B=0.5T in Fig ED1) and avoiding accidental double dots in one junction. All above strongly favor using InSb, which has a smaller effective mass and larger g-factor compared to similar alternatives such as InAs.

9. Line 175: The authors state that for B//Bso the spin is a good quantum number. I am a bit puzzled by that statement. Once there is spin-orbit one has a pseudospin, i.e. not a real spin any more. Do the authors base their statement on the fact that they did not see an avoided crossing in the excitation spectra in ED3 when B//Bso?

We believe the statement that spin is a good quantum number when B // B_{SO} is correct, since there are no non-commuting Pauli spin terms in the Hamiltonian. We have improved the text to reflect that the direction of B_{SO} is inferred from the observed spin conservation in CAR and ECT. We refer to page 3 of this document for a more detailed discussion of the determination of BSO in reply to referee 1.

In addition and again about Fig. ED3: at what VLD and VRD values have the measurements been taken? Are the voltages used to define the QDs similar to those used for figures 3 and 4?

The gates settings used in Fig 3 and Fig 4 are different than those used to measure Fig ED3 and the original Fig ED8. Unfortunately, during the cool down and after measurement of Fig 3 and Fig 4, the device experienced an electric shock that altered the potential landscape on the right side. While the device behaviour remained qualitatively similar, the gate voltages needed to reach approximately the same QD occupation changed significantly. We thus could only take the value of B_{SO} of QD measured in Fig ED3 as a rough estimation for the number relevant to Fig 3 and 4. We clarify this in the figure caption of Fig ED3:

"The measurements shown here were conducted using different QD orbitals than those used in Fig. 3 and Fig. 4. The obtained magnitude of the SOC should be taken as an estimate rather than a precise value."

10. I am surprised by the statement that the hybrid section has significantly larger SOC. Also by the fact that their sample with and without Pt shows qualitatively the same behavior. The authors state in their recent arXiv (Spin-mixing enhanced proximity effect in aluminum-based superconductor-semiconductor hybrids) "These heavy atoms introduce spin-orbit scattering, which prevents spins from being polarized by an external magnetic field". So should the two samples (device A and device B) not behave differently?

The referee raises two questions: why the SOC in hybrid segment is larger than that of the QDs themselves, and what is the role of the Pt sub-layer on top of the Al superconducting contact.

To answer the second question, the addition of Pt atoms is indeed established to result in spin mixing *within the metallic superconductor*, thus quenching the Zeeman effect and increasing its critical field. This effect is well known and, as the referee points out, we apply it to our Al shell on the nanowire in another manuscript (G. Mazur et al, *Advanced Materials,* 202202034). There, we argue that the spin-mixing effect on top of the Al shell does not leak into the proximitized semiconductor by showing Zeeman-split Andreev bound states with a large *g*-factor. In fact, since the proximitized semiconducting properties depend strongly on microscopic details such as interface and disorder scattering in the two materials, advanced theory models would be required for a detailed quantitative understanding. Moreover, one of our motivations to fabricate and measure Device B was to verify that the spin-mixing effect of Pt can be disregarded in our interpretation of the experiment. The qualitative agreement between these two devices (and others not discussed here) confirms the assertion that spin in InSb is not mixed by Pt on the top surface of Al.

The magnitude of the SOC in the hybrid is not known a priori. On the one hand, the hybrid segment is longer than the size of the QD which leads to more SOC-induced spin precession. On the other hand, the hybridization of the semiconducting state with those of Al might lead to a lower effective Rashba coefficient. Yet another complicating factor is that the electric field in the hybrid segment at a given gate voltage heavily depends on the material band offset (which can lead to a steep voltage drop at the interface) and is not conclusively known. Whether all elements combine to produce stronger or weaker SOC cannot be predicted given the little we know regarding microscopic details. However, we can phenomenologically separate the contribution of SOC-perturbated QD eigenstates from the total anisotropy, leaving the rest attributed to the segment in between QDs. The calculation shows that the measured SOC in the QDs themselves is insufficient to produce the amount of observed triplet component. This amount, instead, is consistent with SOC in the hybrid with a magnitude of 0.11 – 0.18 eV A, in agreement with other values reported in the literature for our system. This is discussed in detail in the caption of Fig ED5 and the supplementary material.

To summarize: we observe an anisotropic equal-spin CAR process, which to the best of our knowledge, can only be realistically attributed to SOC. That the order parameter in the hybrid segment contains a spin-triplet component is one plausible interpretation but is still undergoing more detailed investigation before it can be conclusively proven. We make the distinction between observation and interpretation better by moving this entire question into the discussion segment. We further elaborate on this point in our reply to referee 3.

11. In order to induce triplet correlations in the two QD, does the spin quantization axis in the two dots need to be the same?

Spin-polarizing the QDs along non-collinear axes, as the referee asks, is in fact one of the two mechanisms to generate triplet correlations between them (the other being SOC). This idea, for example, is explicitly illustrated in

one proposal for the creation of a Kitaev chain (Leijnse, et al., PRB 86, 134528 (2012)), although difficult to implement in our devices. This is also mentioned in the supplementary material in the section *SOC in QDs*.

12. It is not clear to me how in the supplementary information they conclude that the expected triplet current at phi=0,180o should be no more than 6%. Can they please explain this further?

The paragraph in which we arrive at this number includes a detailed description, the outline of which we summarize here. We can learn the spin composition of a QD level, $|\psi\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle$, via QD excited-state spectroscopy (Fig ED3). The spin-overlap $|\langle\psi_L|\psi_R\rangle|^2$ of two such QD levels are taken as the probability of transition between the two. The 6% figure is an order-of-magnitude estimation of the spin-flipping/spin-conserving tunneling rates by plugging in the worst-case numbers measured.

13. Fig ED1: Can the authors explain how they tune up the devices?

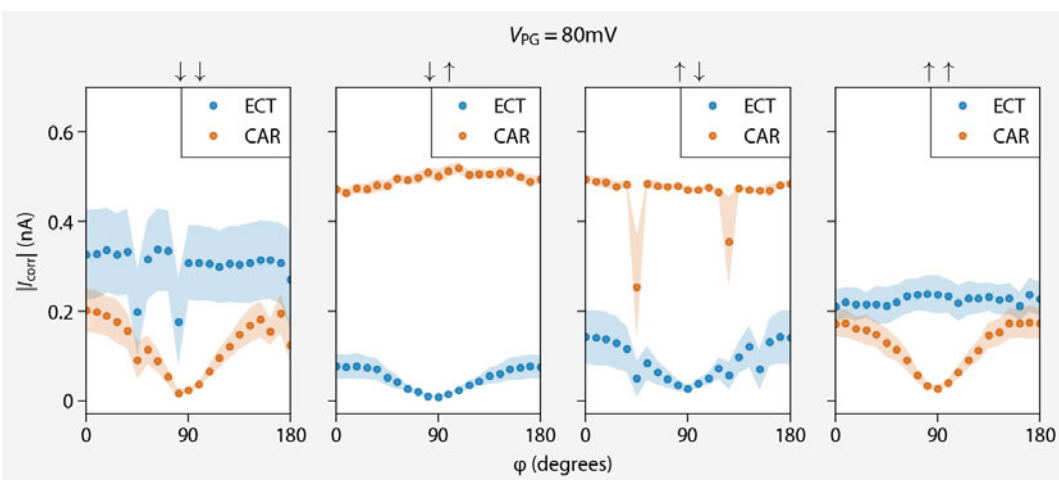
The question of tuning up our devices is important and was discussed in reply to question 1 of the referee. We added a Methods section to describe our tuning protocol.

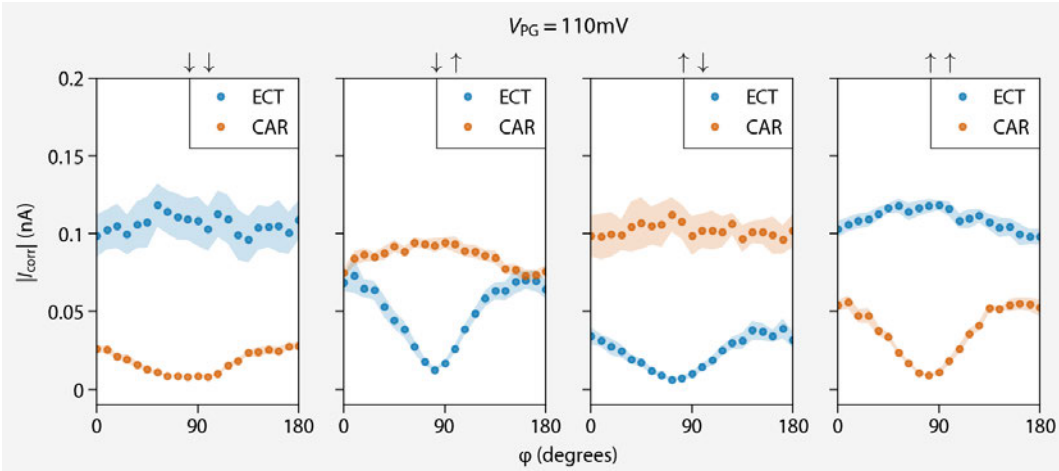
14. Fig ED7d. At which VLD has this plot been taken?

Fig ED7d was taken at V_{LD} = 0.357 mV. We add this value to the figure caption.

15. The authors have been focusing on 4 configurations in their current versus VRD and VLD plots. Do the devices show similar behavior for different gate voltage configurations? Is the CAR process indicating triplet superconductivity observed for a wide range of gate voltages?

The behavior we report on in this work is generic and is observed for wide range of gates across multiple devices. Below, we show two more examples taken at different values of V_{PG} on Device A showing similar behavior – presence of equal-spin CAR and opposite-spin ECT that are suppressed along a special *B* angle.




V_{PG} = 110mV
↓↓
↓↑
↑↓
↑↑
ECT
CAR
$|I_{corr}|$ (nA)
0.2
0.15
0.1
0.05
0
0
90
180
φ (degrees)

Referee #3 (Remarks to the Author):

The manuscript by Wang et al. reports on Cooper pair splitting experiments with spin sensitivity. Two electrons of a Cooper pair have opposite spin (for ordinary s-wave superconductors) and by splitting them to two quantum dots have been theoretically proposed to yield spatially separated entangled electrons since long. While the manuscript doesn't report on entanglement, it achieves an important milestone by demonstrating a high fidelity spin selectivity and spin selection rules that apply for the splitting process. In particular the spin blockade results of Figs. 3 and 4 in the context of split pairs are conceptually new and the results presented around them are clear and sound. I find this content to fit well to the scope of the journal.

We thank the referee for the positive assessment of our work.

In addition to the above results, the authors use theory arguments to claim that the superconductor-semiconductor hybrid section that they use would have triplet pairing in it. Their argumentation is based essentially on the fact that the length of the quantum dots is shorter that the superconductor-semiconductor hybrid section and hence the spin orbit coupling (SOC) in the hybrid section would to be stronger than in the quantum dot. While this would true, the argumentation does not take into account that during the tunneling process, the electron moves much longer distance than the quantum dot size, and as this distance is comparable or larger than the hybrid segment, the SOC during the tunneling event potentially have the biggest impact to flip the spin. Also, as the authors correctly point out, the fact that the electronic states in the hybrid segment need to be at least partially in the aluminium segment, the SOC in the hybrid part is suppressed from the strong SOC of the bare InSb.

Our main observation is anisotropic equal-spin CAR and opposite-spin ECT. These processes are suppressed along a given B direction, generally perpendicular to the axis of the nanowire, and reach a maximum when B is applied along the nanowire direction. We interpret this as the presence of triplet CAR coupling between the QDs, which is generated by spin-precession due to SOC. We stand by this interpretation.

We make another step in our interpretation, which is to argue that the triplet pairing arises predominantly from the hybrid segment of the device instead of the QDs. We base this interpretation on assessment of the intrinsic spin-orbit coupling in the QDs and argue that the measured value is insufficient to generate the amount of triplet component we observe. Therefore, spin precession must have occurred during the journey from left to right QD.

On this point, the referee makes a clearer distinction than we did between the tunneling process through the bare semiconductor barriers and the Al-covered segment as the reason for spin precession. To accommodate this, we modify our manuscript to separate the observed presence of triplet CAR component and discussion of the possible microscopic mechanisms for such pairing. We leave open the question of where SOC precisely resides. Compelling evidence for the role of the Al/InSb bulk in the spin precession would require showing the triplet component evolving in response to V_{PG}. Although we have preliminary data showing that indeed V_{PG} significantly influences the spin precession, we will address this in a future publication after we obtain more complete data to support such claims.

Regardless of the microscopic origin of SOC precessions, from the QDs' point of view, the entire middle section of the device generates triplet superconducting pairing, whether the active part is the entire barrier–Al/InSb–barrier heterostructure as the referee proposes (similar in spirit to SFS junctions) or only inside the Al/InSb bulk. Either case allows us to model the hopping process between QDs phenomenologically as those in a Kitaev chain Hamiltonian.

We want to emphasize that our measurements do no hint at a topological phase inside the hybrid: inducing triplet pairing between the dots only requires spin-orbit, but does not require the hybrid region to have a purely p-wave order parameter. Our previous manuscript was ill-phrased in this regard. We have been more careful now in emphasizing that the triplet pairing is between the quantum dots and that we consider a triplet pairing *component* (in addition to a singlet component) in the hybrid segment a plausible explanation.

We modify the first part of our discussion segment (lines 545-588):

"The oscillating CAR signals in Fig. 4g for the two equal spin configurations (most left and most right panels) are the central results of this work. The presence of the anomalous equal-spin CAR signal shows how non-collinear B and B_{SO} leads to unconventional spin pairing. Below we discuss possible microscopic scenarios giving rise to SOC-induced spin precession. InSb nanowires have both Rashba- and Dresselhaus-type SOC.

Both terms are linear in the momentum along the nanowire axis and their addition gives an effective spin-orbit term in a direction generally perpendicular to the nanowire axis [40]. Such SOC exists in our InSb-based QDs and can lead to nominally ↑ QD eigenstates having a small ↓ component [43]. In Fig. ED3 and Supplementary Material, we quantify this effect and argue that the opposite-spin admixture is too small to explain the measured amplitude of the ECT and CAR anisotropy.

The superconducting pairing in the hybrid segment itself is predicted to hold a triplet component due to SOC as well [44]. The shape and amplitude of our observed oscillations allow comparison with a theory adopting this assumption [32], resulting in an estimated spin-orbit strength in the hybrid section between 0.11 and 0.18 eV ·˚A for Device A and 0.05 to 0.07 eV ·˚A for Device B (see Fig. ED5). This estimation agrees with reported values in the literature [45, 46]. While the existence of triplet pairing component in the hybrid is thus consistent with our results, it is not the only possible explanation. During the tunneling process between the QDs, the electrons traverse through a bare InSb segment, whose SOC could also result in spin precession [38, 40]. Both scenarios, however, support an interpretation of spin-triplet superconducting coupling between the QDs necessary for construction of a Kitaev chain [8]. "

> Also, the manuscript does not present any direct experimental evidence about the triplet pairing but rather probes the spin directions outside of the superconductor. For a compelling evidence of the triplet p-wave pairing, one should present results about the formation of the Kitaeev chain, not done in this work.

Indeed, the formation of Kitaev requires the presence of coherent triplet pairing between the QDs. We show precisely this in a parallel manuscript (arXiv:2206.08045) with Device A using different gate settings.

> Because of these reasons, I find the claims about the triplet pairing non-conclusive and speculative and they should be removed from the manuscript before publishing. In particular, I find the claim "Our results demonstrate controllable detection of singlet and triplet pairing" in the abstract misleading. As stated above, I find the manuscript content excellent but such overstatements about the p-wave pairing should be removed.

As in the answer to the first comment from this referee, we can argue with a large degree of certainty that we detect a triplet component CAR between the two QDs, but cannot verify that we detect triplet p-wave superconductivity in the bulk hybrid alone without the barrier-hybrid-barrier heterostructure. We have modified such claims to be more precise. Specifically, the sentence the referee mentions is modified to **"Our results demonstrate controllable detection of singlet and triplet pairing between the quantum dots. "**

> More minor comments/questions on the manuscript content:
>
> 1) Direction of the B_SO: I don't see any arguments why the direction of the B_SO field should be along the y-axis of Fig. 4a. What is the argumentation for this particular direction? This would be worth clarifying. I presume the nanowire may land at an arbitrary angle to the substrate and hence the B_SO can be at any angle in the y-z plane. It is also claimed that the device B has a 30 degree angle between the substrate surface and B_SO. How was this estimate obtained? It would be also good if the authors could comment on why the Dresselhaus contribution is neglected in the analysis.

The shadow-wall lithography technique we employ to fabricate these devices (see Heedt et al.) is deterministic: bare nanowires are placed in a well-defined position and orientation, and an Al layer is only then evaporated at a predesignated angle with respect to surface. However, the direction of B_{SO} is not known a prioi, and is influenced by the microscopic potential landscape and applied electric fields. We argue that the measurements we conduct in fact reveals the direction of B_{SO}. When the magnetic field is applied along B_{SO}, spin is conserved and we do not observe equal-spin CAR or opposite-spin ECT. Thus we conclude that the direction in which spin is conserved is the direction of B_{SO}. We recognize that this point was presented in a confusing manner and clarify it in our modified manuscript (404-472):

"We first apply B = By = 100 mT, in the plane of the substrate and perpendicular to the nanowire. The four panels in Fig. 4c present I_{corr} measured at four bias polarity combinations, selecting either CAR or ECT conditions. The upper left panel also shows the lowest-energy spin combinations. I_{corr} vanishes for ↑↑ and ↓↓ with CAR biases −− and ++, and for ↑↓ and ↓↑ with ECT biases +− and −+. The observation of spin conservation suggests spin is a good quantum number. Thus, any spin-orbit field in the InSb nanowire, B_{SO} (including both possible Rashba and Dresselhaus SOC), must be parallel, or nearly parallel, to By. In this

case, the superconducting pairing between the QDs is singlet. We note that the B_{SO} direction as measured by suppression of equal-spin CAR/opposite-spin ECT depends on gate settings and the device used (e.g., Fig. ED6). We have measured directions within 20° of being perpendicular to the nanowire axis but its angle with the substrate plane can range from 0 to 60°. This observation is consistent with the expectation of BSO being perpendicular to the nanowire axis for both Rashba and Dresselhaus SOC [38–40].”

Following the referee’s suggestion we indeed notice that we cannot distinguish in our system between Rashba and Dresselhaus spin-orbit couplings, as both give a similar contribution to the transport. We modify our text to discuss both types of spin-orbit coupling by adding the following text (555-562): **“InSb nanowires have both Rashba- and Dresselhaus-type SOC. Both terms are linear in the momentum along the nanowire axis and their addition gives an effective spin-orbit term in a direction generally perpendicular to the nanowire axis [40].”**

> 2) Caption of Fig 4, line 156: "applied B direction defined as the spin quantization axis." and main text line 170: "We define the applied B direction as our spin direction axis" are somewhat sloppy formulations. It's rather that you apply a magnetic field to a specific direction and as a result, the magnetic field sets the spin quantization axis. It's a matter of what field you apply in the experiment, not a matter of what definition you take for the applied magnetic field. I suggest to reformulate these sentences accordingly.

We follow the referee’s suggestion and reformulate these sentences.

> 3) Fig. ED2 and text around line 414: the background current I_BG seems to be taken in Coulomb blockaded regime where charging energy cost suppresses the transport. A more truthful estimate of the background level could be taken inside the transport windows at the anti-diagonal areas. Inside the transport window, sequential single-electron tunneling is possible without a charging energy cost, as well as possibly other competing processes, which may lead to lower values of the splitting estimates than what is reported now.

In fact, $I_{L,BG}$ is taken as the average current when V_{RD} is off-resonance and V_{LD} is *within* the bias window. Therefore, the current is not suppressed by Coulomb blockade, but rather is only suppressed due to the superconducting gap. The same procedure is applied for $I_{R,BG}$. We now explicitly mention that V_{LD} is on resonance. Previous Cooper-pair splitter studies adopted the same definition and we would like to keep the metric consistent to enable comparison.

> 4) Fig. ED6 panels e-p: These would be easier to read if they would be arranged into a 2x2 matrices similarly as the data in Figs. 3c and Fig. 4c and f. With the current organization of the panels, it is very laborious to assess if the data follows the correct structure in line with what is presented in the main manuscript figures.

We accept the suggestion made by the referee and improved the plotting.

> 5) Fig. ED8: Magnetic field magnitude and voltage settings are not given. When applying perpendicular field as done here, I would be expecting the superconductivity to become suppressed and influence on the analysis. Also, what relevance does the 18.3 degree angle have? I don't see what it the point of this fitted value.

Following the modification made to the introduction of B_{SO} in the main text, we feel that Fig ED8 does not serve an important purpose and we replace it with a spin correlation analysis suggested by referee 1 in the modified manuscript.

> 6) References: Since this study is about new spin blockade effect, it would be good to include in the discussion the pioneering work of K. Ono et al, Science 297, 1313 (2002) about the Pauli spin blockade. Also the more recent works A. Hofmann et al, Phys. Rev. Lett. 119, 176807 (2017) and Ji-Yin Wang et al. Nano Letters 18, 4741 (2018) that discusses the SOC and its anisotropy in spin blockade would probably be useful references for the B_SO argumentation and discussion.

We thank the referee for pointing out these references. They, and in particular Hofmann et al., are very relevant for this work, both as foundations for the anisotropic effect of B_{SO} and to show the importance of the tunneling process in this phenomenon. We include them in the relevant places of our modified manuscript.

Reviewer Reports on the First Revision:

Referees' comments:

Referee #1 (Remarks to the Author):

The resubmission of the paper by Wang and co-workers incorporates the comments by the reviewers in a way that makes the text more accessible, which was my main criticism previously. The data is now explained in a more transparent way, and in my opinion with the revisions the text has now reached the quality suitable for publication in this journal. The experimental data is of high quality and of high significance, and the analysis is now put into an appropriate framework. My further comments remain minor.

With Figure ED8 the authors followed my recommendation of replacing their previous analysis by the shown spin correlators. I find this figure indeed more useful, and in fact very instructive. The way the spin correlation is defined should have a reference though, as it is based on constructing a spin correlator through conductance measurements. This is in contrast with the older literature on Bell tests in nanostructures that makes such constructions through noise correlators. The two theory papers I cited in my previous report are distinct in that they use spin filtered conductances for this reconstruction instead, and to my knowledge this was only introduced with the cited Phys. Rev. Lett. In the figure caption it should moreover be specified over what the 'max' is taken to define CAR.

The use of SOC as an effective B field has now a better justification, although I still think it appears rather abruptly and focuses directly on in what directions spin remains a good quantum number. For a reader knowing SOC this indeed provides all necessary information though. But the authors may want to consider if they don't want to add still a few more introductory words.

Why do the two panels in Figure ED9 have different colour schemes? I find it particularly puzzling that the left panel has black and the right white for g=0.

In the supplement, the authors may want to give a reference to obtaining Equations (1) and (2), and to the KWANT package.

Many references are incomplete and miss year or journal information. Attention should also be paid for capitalising journal titles and names such as Kitaev or Josephson.

Referee #2 (Remarks to the Author):

I thank the authors for very carefully replying to all raised points. I feel that the revised manuscript has further improved. I am in favor of publication. Just one minor point: I would suggest to include the figures shown on page 10 and 11 of the reply in the supplementary information.

Referee #3 (Remarks to the Author):

The authors have improved the manuscript more or less adequately along to what I was requesting. Most importantly, the high quality data and impressive results stand well the criticism.

I have just one remaining remark on the main concern I expressed in the first report. This issue deals with what the authors called/call "triplet pairing", i.e. the fact that the authors are probing the spins in the quantum dots, not the spins of the Cooper pairs in the superconductor. While the authors made considerable improvements to the presentation to make this point more evident, I find the presentation still highly likely to be confusing for a non-expert audience. The authors use varying technical jargon terms in the manuscript such as "triplet pairing between the quantum dots", "the Cooper pair spin pairing", "the superconducting pairing between the QDs" and "spin pairing between QDs". These will likely be confusing for a non-expert reared as some of the statements refer to the quantum dots and some to the Cooper pairs or superconductivity. With this mix of terms, it's likely hard for the general audience to discriminate if the work is about coupling the spins in the two quantum dots (which is the case well established by the presented work) or if it is about the spins of the Cooper pairs of the superconductor (for which the work gives a couple of different scenarios but does not achieve to give a definite answer). In particular, the statement "Under these conditions, we use QD spin filters to examine the Cooper pair spin pairing." remaining in the manuscript on lines 392 - 394 gives a misleading impression that the authors are probing the spins of the Cooper pairs, which is not the case.

To avoid confusion, I still restate my earlier recommendation to use term(s) consistently referring to the QD spins (and avoiding confusing terms referring to the Cooper pairs and superconductivity) when making statements about the achievements of the presented work.

Author Rebuttals to First Revision:

We wish to thank all three referees for their careful reading of our report and for the constructive comments allowing further improvement in the quality of manuscript. Below we give a point-by-point response to all comments made by the referees.

Referees' comments:

Referee #1 (Remarks to the Author):

The resubmission of the paper by Wang and co-workers incorporates the comments by the reviewers in a way that makes the text more accessible, which was my main criticism previously. The data is now explained in a more transparent way, and in my opinion with the revisions the text has now reached the quality suitable for publication in this journal. The experimental data is of high quality and of high significance, and the analysis is now put into an appropriate framework. My further comments remain minor.

We thank the referee for reviewing our manuscript for the second time and for their positive assessment of our work.

With Figure ED8 the authors followed my recommendation of replacing their previous analysis by the shown spin correlators. I find this figure indeed more useful, and in fact very instructive. The way the spin correlation is defined should have a reference though, as it is based on constructing a spin correlator through conductance measurements. This is in contrast with the older literature on Bell tests in nanostructures that makes such constructions through noise correlators. The two theory papers I cited in my previous report are distinct in that they use spin filtered conductances for this reconstruction instead, and to my knowledge this was only introduced with the cited Phys. Rev. Lett. In the figure caption it should moreover be specified over what the 'max' is taken to define CAR.

We add the required citations to explain that the correlation is constructed using conductance measurements. We further note in the figure caption of ED8 that the maximum of Icorr is taken as defined in the explanation of Fig. 4g of the main text. To avoid confusion and unify the notation, we remove the label 'max' in the figure caption.

The use of SOC as an effective B field has now a better justification, although I still think it appears rather abruptly and focuses directly on in what directions spin remains a good quantum number. For a reader knowing SOC this indeed provides all necessary information though. But the authors may want to consider if they don't want to add still a few more introductory words.

We now explain in the introduction how spin orbit coupling alters the system. We add the following sentence:

"For instance, spin-orbit (SOC) coupling can rotate an opposite spins configuration into an equal-spin pair."

Why do the two panels in Figure ED9 have different colour schemes? I find it particularly puzzling that the left panel has black and the right white for g=0.

ED9 shows standard 3-terminal conductance measurements. The choice of color scales is driven by present convention in the field which we would like to keep following. This way, the local conductance, being only positive, uses a sequential color scale. The non-local conductance, being both positive and negative, is better visualized using a diverging color scheme centered around zero. The contrast between the two color scales also serves as a visual hint to the reader on whether a plot is local or nonlocal conductance.

In the supplement, the authors may want to give a reference to obtaining Equations (1) and (2), and to the KWANT package.

Equations 1 and 2 of the supplement are taken from reference 5 introduced in the preceding paragraph. We now include this reference immediately next to the equations. We have also added the reference to the KWANT package.

Many references are incomplete and miss year or journal information. Attention should also be paid for capitalising journal titles and names such as Kitaev or Josephson.

We corrected all the bibliographic information.

Referee #2 (Remarks to the Author):

I thank the authors for very carefully replying to all raised points. I feel that the revised manuscript has further improved. I am in favor of publication. Just one minor point: I would suggest to include the figures shown on page 10 and 11 of the reply in the supplementary information.

We thank the referee for reviewing our manuscript for the second time and for their positive assessment of our work. We would like to respectfully decline the suggestion to add the figures appearing in our response to the paper for the following reasons:

1. The figure shown in page 10 of our response is a main figure in a publication we are currently preparing, and thus is considered confidential. It will hopefully appear online soon.
2. The data shown in page 11 was observed in the past in the context of the study of Andreev bound states in semiconductors. Since we are already at the limit of the number of extended data items allowed by Nature, we cannot directly add this figure. We hope that interested readers will read the peer-review document where this figure will be available.

Referee #3 (Remarks to the Author):

The authors have improved the manuscript more or less adequately along to what I was requesting. Most importantly, the high quality data and impressive results stand well the criticism.

We thank the referee for reviewing our manuscript for the second time and for their positive assessment of our work.

I have just one remaining remark on the main concern I expressed in the first report. This issue deals with what the authors called/call "triplet pairing", i.e. the fact that the authors are probing the spins in the quantum dots, not the spins of the Cooper pairs in the superconductor. While the authors made considerable improvements to the presentation to make this point more evident, I find the presentation still highly likely to be confusing for a non-expert audience. The authors use varying technical jargon terms in the manuscript such as "triplet pairing between the quantum dots", "the Cooper pair spin pairing", "the superconducting pairing between the QDs" and "spin pairing between QDs". These will likely be confusing for a non-expert reared as some of the statements refer to the quantum dots and some to the Cooper pairs or superconductivity. With this mix of terms, it's likely hard for the general audience to discriminate if the work is about coupling the spins in the two quantum dots (which is the case well established by the presented work) or if it is about the spins of the Cooper pairs of the superconductor (for which the work gives a couple of different scenarios but does not achieve to give a definite answer). In particular, the statement "Under these conditions, we use QD spin filters to examine the Cooper pair spin pairing." remaining in the manuscript on lines 392 - 394 gives a misleading impression that the authors are probing the spins of the Cooper pairs, which is not the case.

To avoid confusion, I still restate my earlier recommendation to use term(s) consistently referring to the QD spins (and avoiding confusing terms referring to the Cooper pairs and superconductivity) when making statements about the achievements of the presented work.

We follow the referee's advice and remove strong claims regarding the Cooper pair spins. This includes the following changes:

1. We add the words "between spin-polarized quantum dots" in the abstract in the context of our measurements of equal-spin pairing.
2. We remove the phrase "we use QD spin filters to examine the Cooper pair spin pairing." (line 392 of the previous version).

3. We changed the previous phrasing “the superconducting pairing between the QDs is singlet” (line 460 of the previous version) to “CAR provides a coupling mechanism only for an opposite spin configuration in the two QDs.”
4. We changed the previous phrasing “The observed CAR coupling for ↑↑ and ↓↓ is interpreted as a measure of the triplet component in the superconducting pairing between the QDs.” (line 500 of the previous version) to “The observed CAR coupling for ↑↑ and ↓↓ is interpreted as a measure of the equal spin coupling between the QDs.”